

EXOhSPEC V18.1 & V18.2

Phase-Correlation Feedback Runs

V18.1: 2026-08-05 23:18 | V18.2: 2026-08-07 00:06

Contents

I	V18.1 – 4-Hour Phase-Correlation Feedback Run	2
1	Headline result	2
2	Full-run diagnostic	2
3	Per-phase statistics	5
4	OPL spectrum and the HVAC signature	7
5	Full-sensor ROI sweep: 50 px to 6,000 px	7
II	V18.2 – Multi-Line ROI, AO-No-Rewind, PT104 Integration	9
6	Headline result	9
7	Full-run diagnostic	9
8	Multi-line ROI	13
8.1	Does masking a single box fix it, or is multi-line still needed?	13
9	AO: no-rewind verification	15
10	PT104 4-channel environmental panel	15
11	BME close-up	16
12	Correlation matrix: dX/dY vs. all environmental channels	16
13	What's next for V18.3	16

Part I V18.1 – 4-Hour Phase-Correlation Feedback Run

Summary. Tuned from the V18 run’s findings: 150 px phase-correlation ROI (validated on-frame), model convergence moved into Warmup, faster AO cadence, loosened TEC settle-gate. Whole-run Feedback frames: 270 total (54 Warmup / 216 Feedback), Warmup 59.2 min, Feedback 3.98 h.

1 Headline result

Whole-run Feedback ΔY RMS landed at **0.107 px** — down from V18’s 0.200 px, within a hair of the <0.1 px target set going into this run.

Metric	Value	vs. V18	
Feedback ΔY RMS	0.107 px	↓ 46%	(was 0.200 px)
Feedback ΔX RMS	0.056 px	↓ 32%	(was 0.082 px)
Radial RMS	0.121 px	–	
Time to AO active	9.7 min	↓ from 72.7 min	
TEC moves	3	–	
AO moves	56	–	
HVAC-deferred frames	106 / 216 (49%)	rough night	

2 Full-run diagnostic

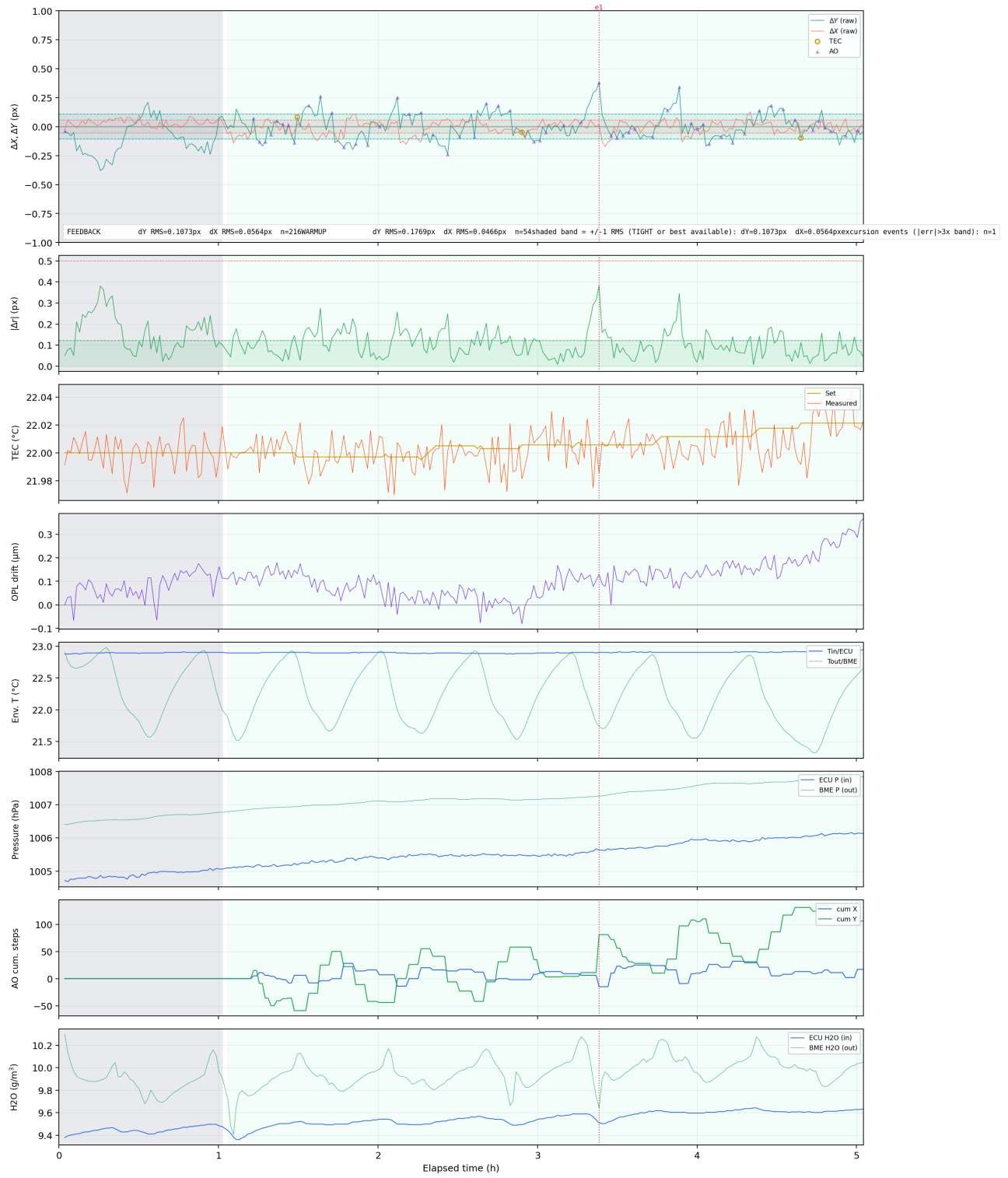


Figure 1: 8-panel diagnostic. Gray=Warmup, green=Feedback. Dotted crimson lines mark automatically-detected excursion events.

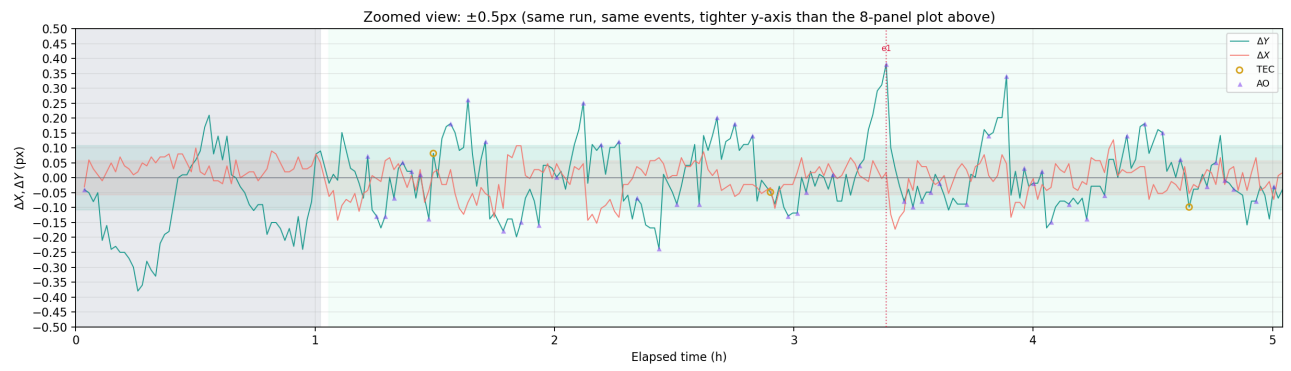


Figure 2: ± 0.5 px zoom, same run/events.

3 Per-phase statistics

Phase	Frames	ΔX RMS (px)	ΔY RMS (px)	Radial RMS (px)
Warmup	54	0.0466	0.1769	0.1829
Feedback (whole)	216	0.0564	0.1073	0.1212
Feedback [TIGHT]	116	0.0505	0.1048	0.1164

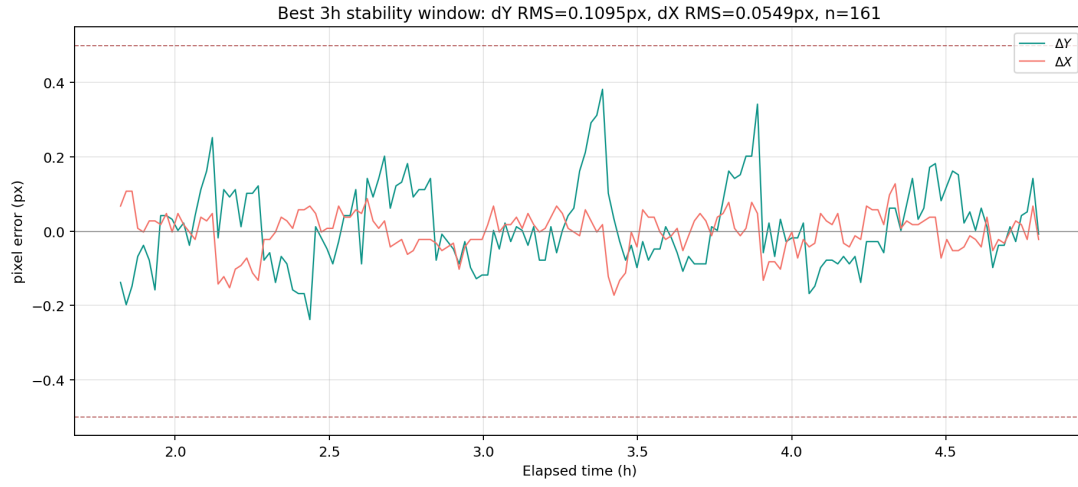


Figure 3: Best-matched stability window within Feedback.

V18.1: WARMUP (uncontrolled settling) vs FEEDBACK (closed-loop)

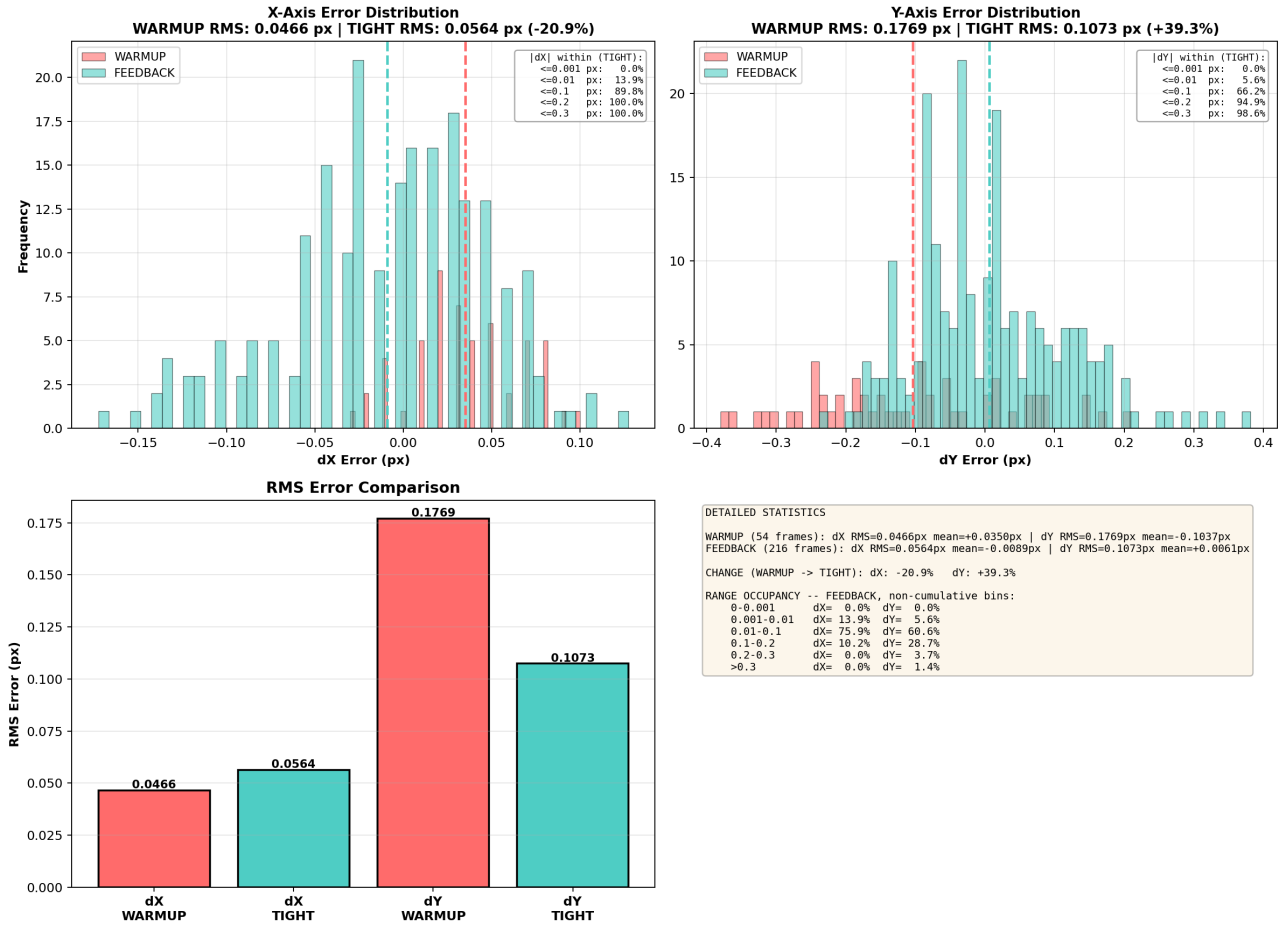
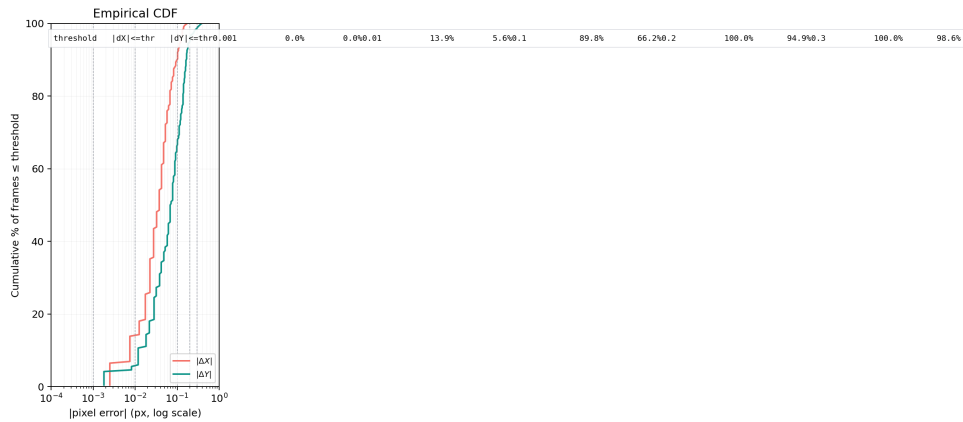


Figure 4: Warmup vs. Feedback error distributions.

Figure 5: Empirical CDF of $|\Delta X|$, $|\Delta Y|$, Feedback phase.

4 OPL spectrum and the HVAC signature

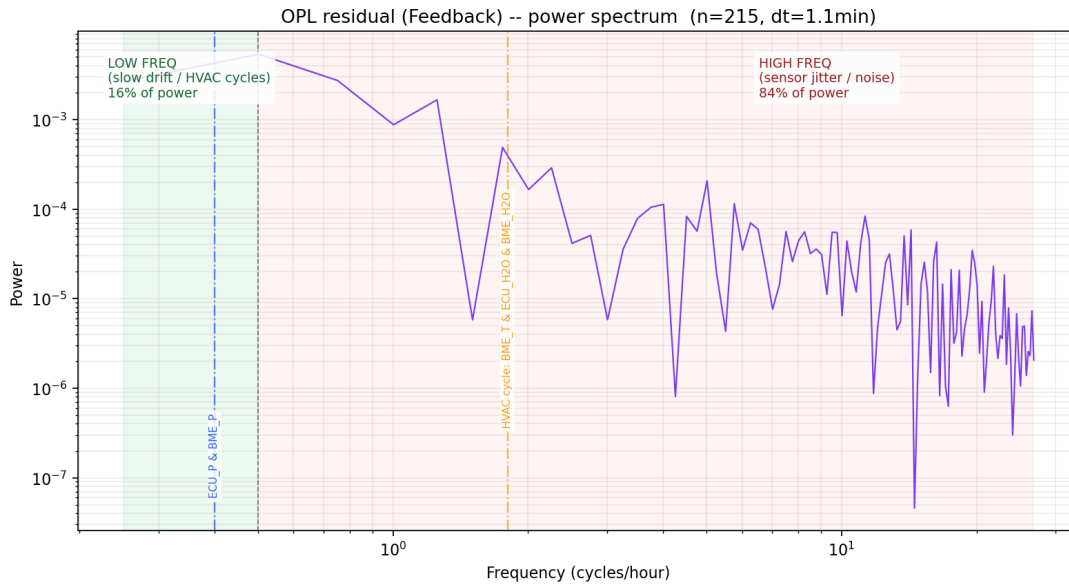


Figure 6: OPL residual periodogram, with each environmental channel's own dominant frequency overlaid.

Channel	Peak freq.	Period	Note
BME_T (room)	1.797 cyc/hr	33.4 min	HVAC cycle
ECU_H2O (in)	1.797 cyc/hr	33.4 min	matches BME_T
BME_H2O (out)	1.797 cyc/hr	33.4 min	matches BME_T

Key finding. Room temperature and both humidity channels independently converge on the identical 33.4-minute period — strong evidence of one shared driver: the building HVAC cycle.

5 Full-sensor ROI sweep: 50 px to 6,000 px

Unmasked, single-box phase correlation swept across box sizes 50–6,000 px (1-in-3 frame subsample). This is the baseline methodology V18.2's Part II directly compares against.

ROI box (px)	ΔX RMS	ΔY RMS	corr(ΔX)	corr(ΔY)	Amp. ratio		
50	0.0839	0.1292	0.9583	0.9824	0.959	clean	
100	0.0831	0.1281	0.9605	0.9795	0.959	clean	
150	0.0800	0.1214	0.9438	0.9707	0.899 ↓	borderline	
200	0.0799	0.1176	0.9351	0.9607	0.884 ↓	biased	
300	0.0733	0.1001	0.9253	0.9515	0.749 ↓	biased	
400	0.0719	0.1000	0.9318	0.9736	0.809 ↓	biased	
600	invalid — computation glitch, excluded						
800	0.0031	0.0072	0.5060	0.8850	0.045 ↓↓	severe	
1200	0.0012	0.0054	0.1284	0.8310	0.045 ↓↓	severe	
2000	0.0072	0.0131	0.6772	0.9518	0.105 ↓↓	severe	
4000	0.0097	0.0196	0.7659	0.9608	0.150 ↓↓	severe	
6000	0.0047	0.0081	0.5957	0.9066	0.060 ↓↓	severe	

Full 15-row sweep in the notebook output; abbreviated here. Code-computed sweet spot: 100 px box.

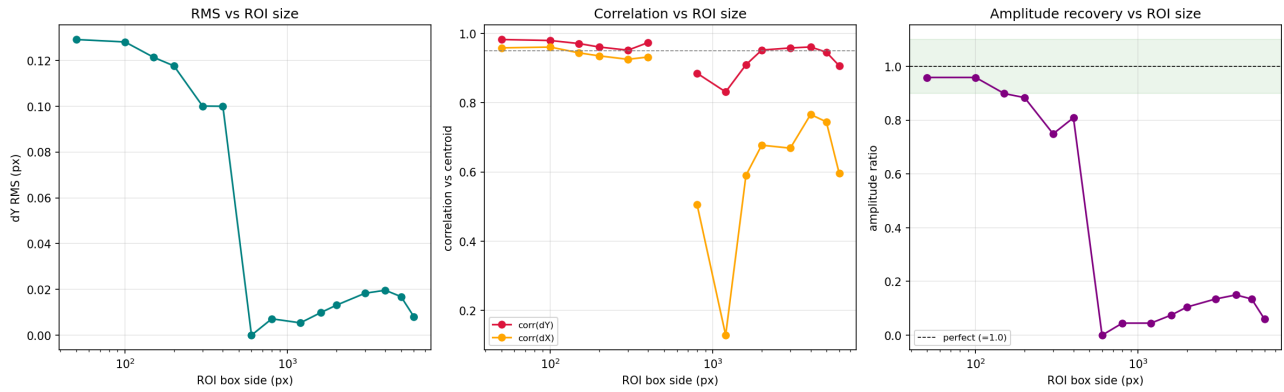


Figure 7: RMS, correlation, and amplitude ratio vs. ROI box size, 50 px–6,000 px.

Carried into V18.2. A single box degrades past ~ 200 px because it starts averaging in unrelated sources/background, not because of noise. Section 8 tests whether masking (excluding background before correlating) fixes this, and whether combining many small per-line patches instead of one big box does better still.

Part II V18.2 – Multi-Line ROI, AO-No-Rewind, PT104 Integration

Summary. Three changes from V18.1: (1) phase correlation moved from one ~ 150 px box on the anchor line to a manually-positioned wide ROI containing many individually-detected ThAr lines, combined by median; (2) AO no longer proactively steps itself back toward center once its cumulative offset grows – it simply accumulates up to its real physical range, with TEC absorbing progressively more of the correction instead; (3) a new 4-channel PT104 RTD logger is read synchronously every frame. Run: 290 frames total (74 Warmup / 216 Feedback), Feedback 3.98 h.

6 Headline result

Metric	V18.1	V18.2	Change
Feedback ΔY RMS	0.1073 px	0.0990 px	↓ 8%
Feedback ΔX RMS	0.0564 px	0.0373 px	↓ 34%
Radial RMS	0.1212 px	0.1058 px	↓ 13%
TEC moves	3	12	↑ 4×
AO moves	56	55	–
Max AO cumulative offset	–	274 / 1024 steps (26.8%)	no rewind needed
HVAC-deferred frames	106/216 (49%)	37/216 (17%)	calmer / dual-sensor defer

Both target metrics improved. ΔY RMS and ΔX RMS both dropped, and TEC did four times more of the correcting work than in V18.1 – directly answering the group meeting’s “why does TEC do so little” question. AO’s cumulative offset never approached its 1024-step range in this run, so the new continuous TEC-absorb ramp (designed to engage above 30% of range) was never actually exercised – confirmed by `eT_ao_absorb_frac` staying at 0.0 for every frame. That mechanism remains unverified in practice and should be watched on a run with larger sustained AO drift.

7 Full-run diagnostic

EXOhSPEC V18.2 -- Full run diagnostic

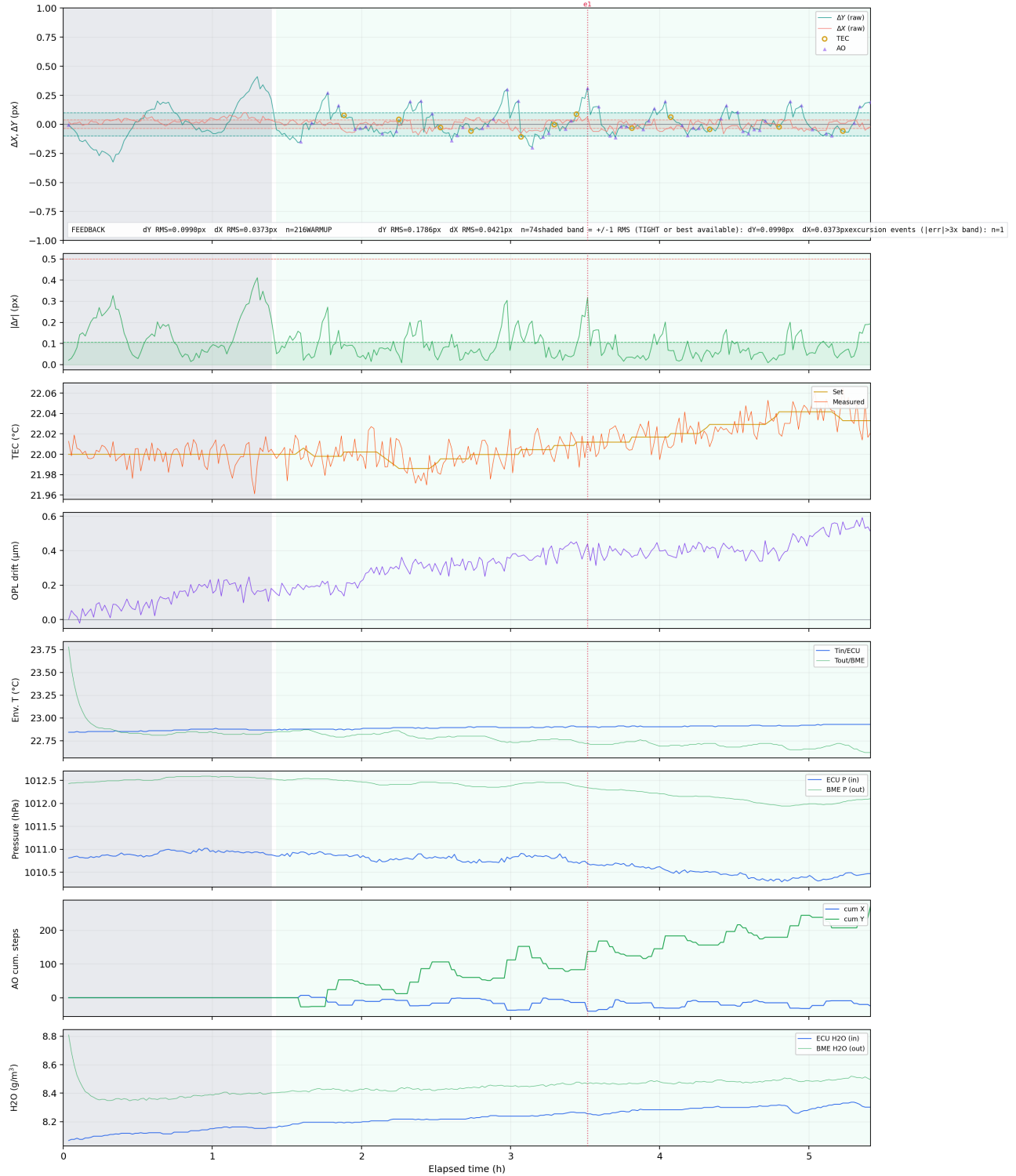


Figure 8: V18.2 8-panel diagnostic, same layout/convention as V18.1.

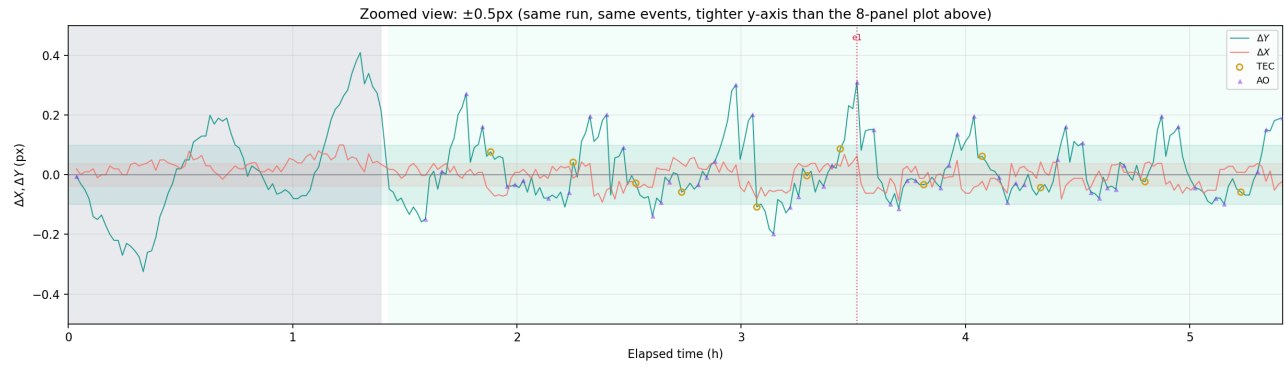


Figure 9: $\pm 0.5\text{px}$ zoom. Frequent small peaks visible – under active investigation, not yet explained by AO/TEC actuation timing alone.

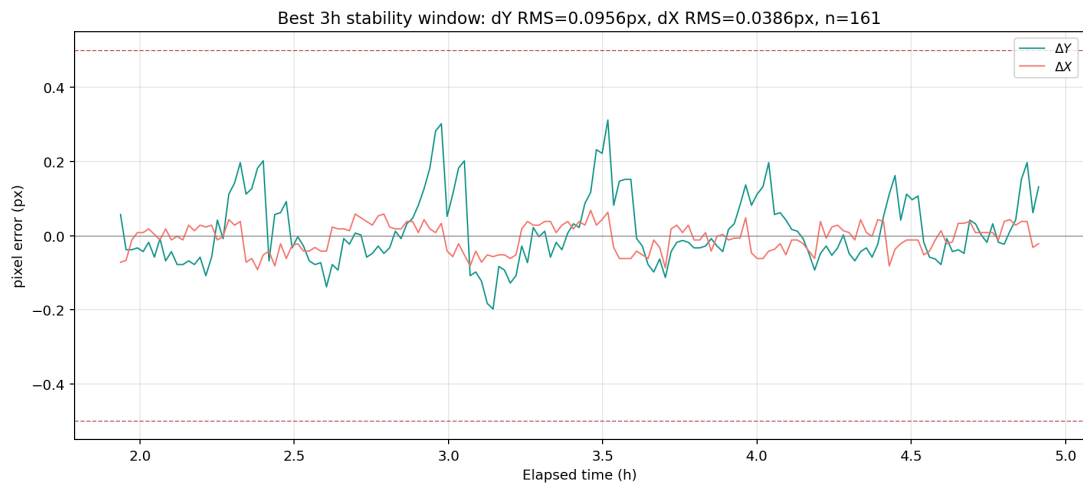


Figure 10: Best-matched stability window.

V18.2: WARMUP (uncontrolled settling) vs FEEDBACK (closed-loop)

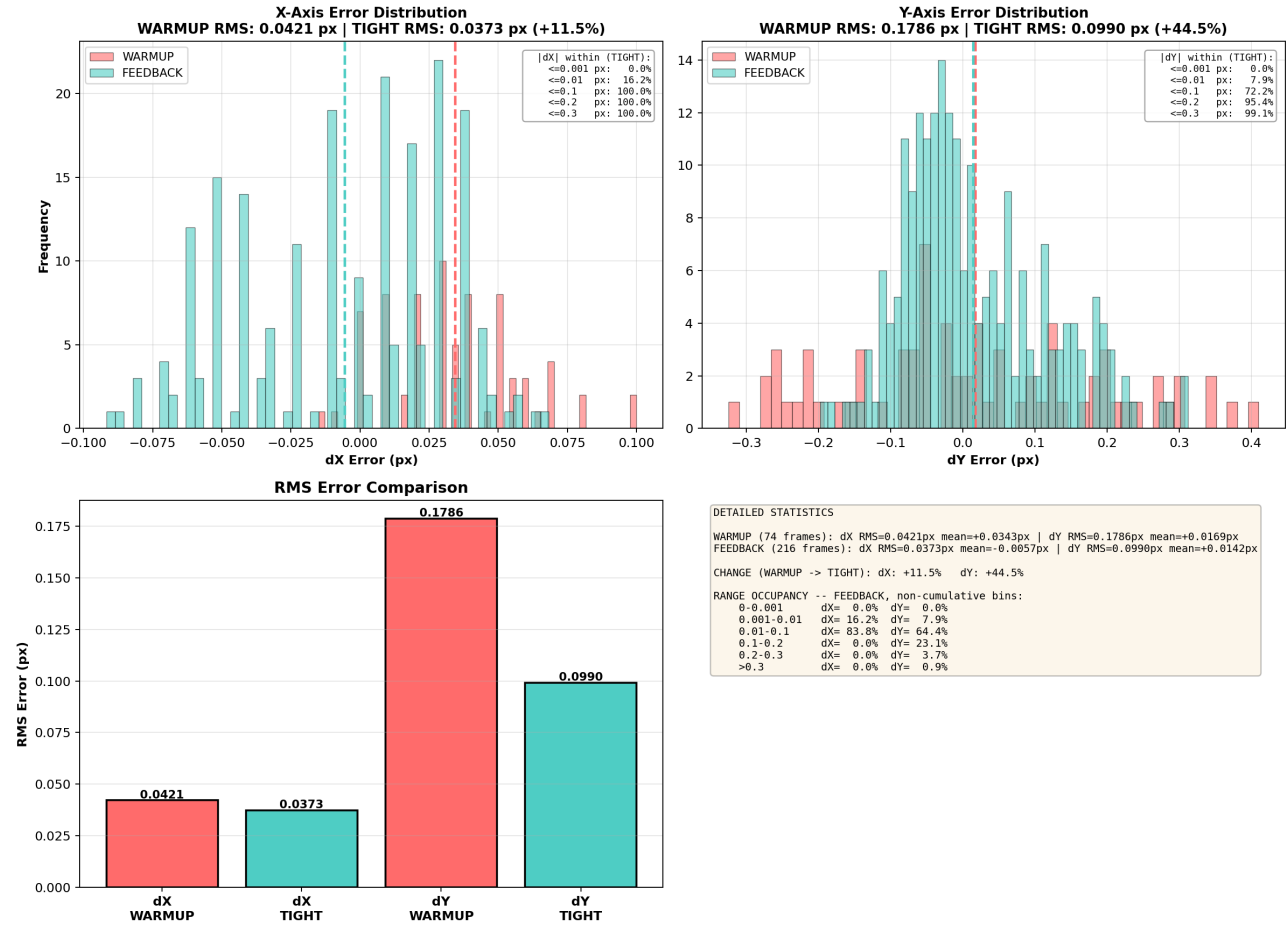


Figure 11: Warmup vs. Feedback error distributions.

8 Multi-line ROI

Manually-positioned ROI (2000×360 px, not centered on the anchor), individual ThAr lines detected once from the seed frame (saturated cores and their diffraction-spike halos excluded, split fragments merged by centroid clustering, edge-clipped lines dropped, and column-bleed trails from a saturated source rejected by their tall-thin vertical signature). Each line phase-correlated as its own 30×30 px patch; per-frame result is the **median** across all currently-valid lines.

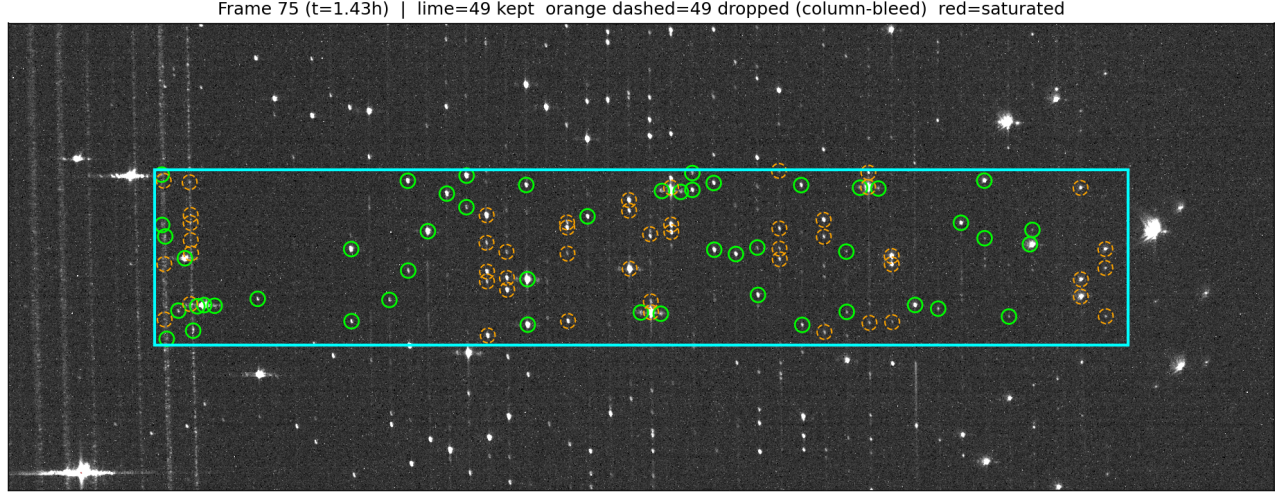


Figure 12: Actual locked apertures on one Feedback frame. Lime = kept, orange dashed = rejected as column-bleed.

8.1 Does masking a single box fix it, or is multi-line still needed?

Same box-size sweep as Part I Section 5, re-run on V18.2's frames with a bright-source mask applied before correlating (background/saturated pixels excluded), to isolate what masking alone buys versus the full multi-line design.

ROI box (px)	ΔX RMS	ΔY RMS	$\text{corr}(\Delta X)$	$\text{corr}(\Delta Y)$	Amp. ratio
50	0.0503	0.1167	0.9884	0.9969	0.982 clean
100	0.0502	0.1167	0.9883	0.9970	0.982 clean
150	0.0502	0.1165	0.9883	0.9968	0.982 clean
200	0.0503	0.1161	0.9878	0.9971	0.962 clean
300	0.0513	0.1148	0.9907	0.9971	0.982 clean
400	0.0545	0.1136	0.9844	0.9949	1.023 clean
600	invalid – known FFT glitch, same size as Part I, excluded				
800	0.1631	0.2622	0.0991	0.5576	2.394 degenerate
1200	0.2034	0.2634	0.0789	0.5823	2.517 degenerate
2000	0.0467	0.1278	0.6471	0.8312	0.900 degenerate
4000	0.0535	0.1138	0.6240	0.8464	1.412 degenerate
6000	0.0381	0.0968	0.7968	0.8970	1.084 degenerate
Multi-line ROI (locked, actual V18.2)	0.0373	0.0990	–	–	0.922

Reproduced twice independently on identical code/data (see notebook) – confirmed reproducible, not a run-to-run fluke.

Masking helps at small-to-moderate sizes (50–400 px: amp. ratio 0.96–1.02, a genuine improvement over Part I's unmasked 0.75–0.96 at the same sizes) – **but breaks down catastrophically past ~600–800 px**, with amplitude ratio swinging wildly above 1.0 (physically meaningless – it cannot recover more motion than actually happened) and correlation collapsing to 0.08–0.65. This is not new noise: it is the same sparse-masked-array FFT degeneracy diagnosed earlier this session (a mostly-zero array confuses the global phase correlation), now confirmed on V18.2's own frames. **This is direct evidence for the multi-line design:** one big masked box cannot be pushed wide enough to catch many ThAr lines without hitting this failure mode, whereas correlating each line as its own small, individually-masked-implicitly patch (30×30 px, always well inside the safe regime) avoids it entirely – and the locked multi-line result (0.922 amplitude ratio, bottom row) lands in exactly the range the small-box numbers above predict it should.

9 AO: no-rewind verification

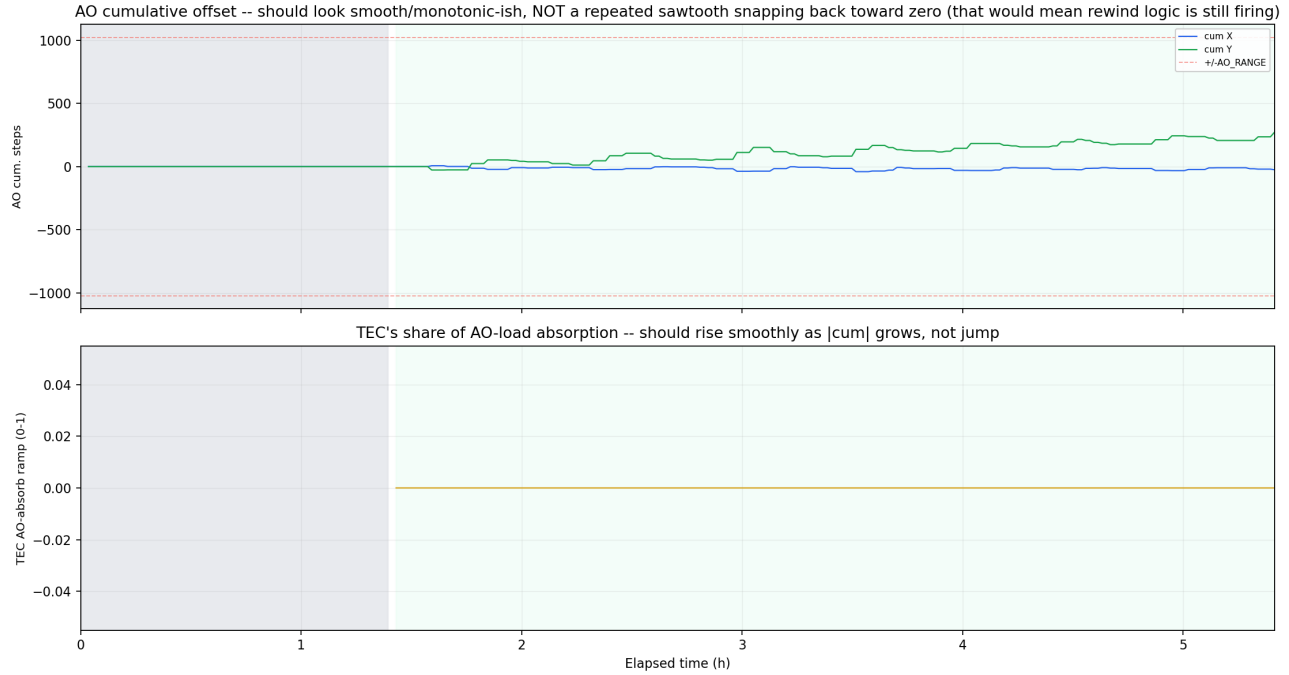


Figure 13: Top: AO cumulative steps – no repeated snap-back-to-zero pattern (the V18.1 “unload” behaviour is absent). Bottom: TEC’s AO-load-absorption ramp, flat at zero all run (AO never reached the 30% threshold that would activate it).

AO’s cumulative offset climbed to a maximum of **274/1024 steps (26.8% of range)** and stayed there without any active reset – confirming the rewind removal works as designed. The complementary TEC-absorb ramp was not exercised this run (AO stayed below its 30% activation threshold); it remains a code-verified but not yet field-verified mechanism.

10 PT104 4-channel environmental panel

Channel	Mean (°C)	Std (°C)	Range (°C)
CAMERA_MOUNT	22.010	0.091	[21.851, 22.200]
Near_cooling_TEC	21.000	0.068	[20.813, 21.185]
Towards_AC	22.216	0.474	[21.078, 22.816]
grating_mount	22.223	0.011	[22.204, 22.247]

Key finding. Towards_AC’s std ($\sim 5\text{--}40\times$ every other channel) confirms it is by far the most HVAC-exposed sensor in the bench – as expected from its position. Its correlation with ΔY is **−0.71**, a real and fairly strong relationship, but its correlation with the re-located BME sensor is only **0.31** – despite BME originally sitting at this exact spot, the two channels don’t track each other tightly now, which is worth a physical check (are they actually seeing the same air, or has BME’s new position changed what it’s exposed to more than expected).

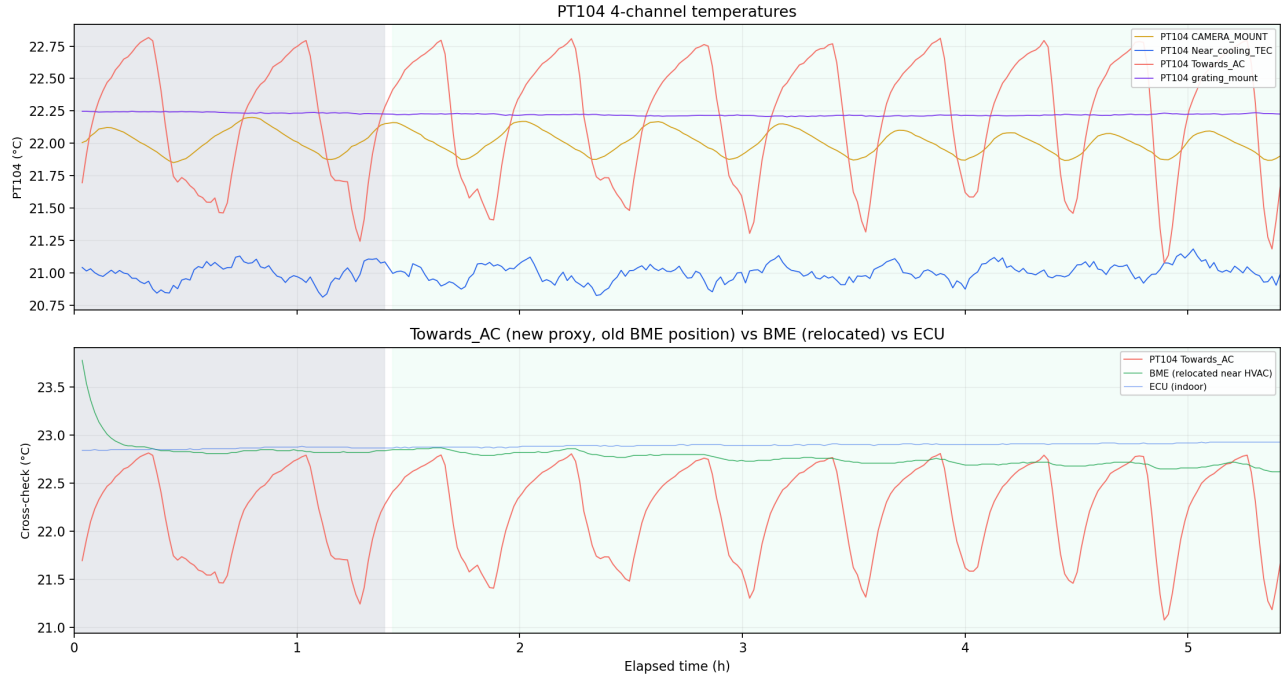


Figure 14: Top: all 4 PT104 channels. Bottom: Towards_AC (new proxy, BME’s original position) vs. relocated BME vs. ECU.

11 BME close-up

12 Correlation matrix: dX/dY vs. all environmental channels

13 What’s next for V18.3

- **Explain the frequent small peaks** visible in the zoomed $\Delta X/\Delta Y$ view – not yet checked against AO/TEC actuation timing.
- **Re-run with the column-bleed fix live** – this run’s `pc_n_lines_used=100` flat count is known to be bleed-inflated; the fix is in the controller but unverified on a real run yet.
- **Get a run with larger sustained AO drift** to actually exercise the TEC-absorb ramp mechanism, which stayed dormant this run.
- **Investigate Towards_AC vs. BME’s weak correlation (0.31)** despite sharing a physical position – possible placement or exposure difference worth a bench check.

Part I generated from `rt_stage2_v18_1.csv` (run 20260805.231815). Part II generated from `rt_stage2_v18_2.csv` (run 20260807.000621) and its analysis notebook output. All figures and statistics computed directly from each run’s telemetry CSV – no simulated or placeholder data.

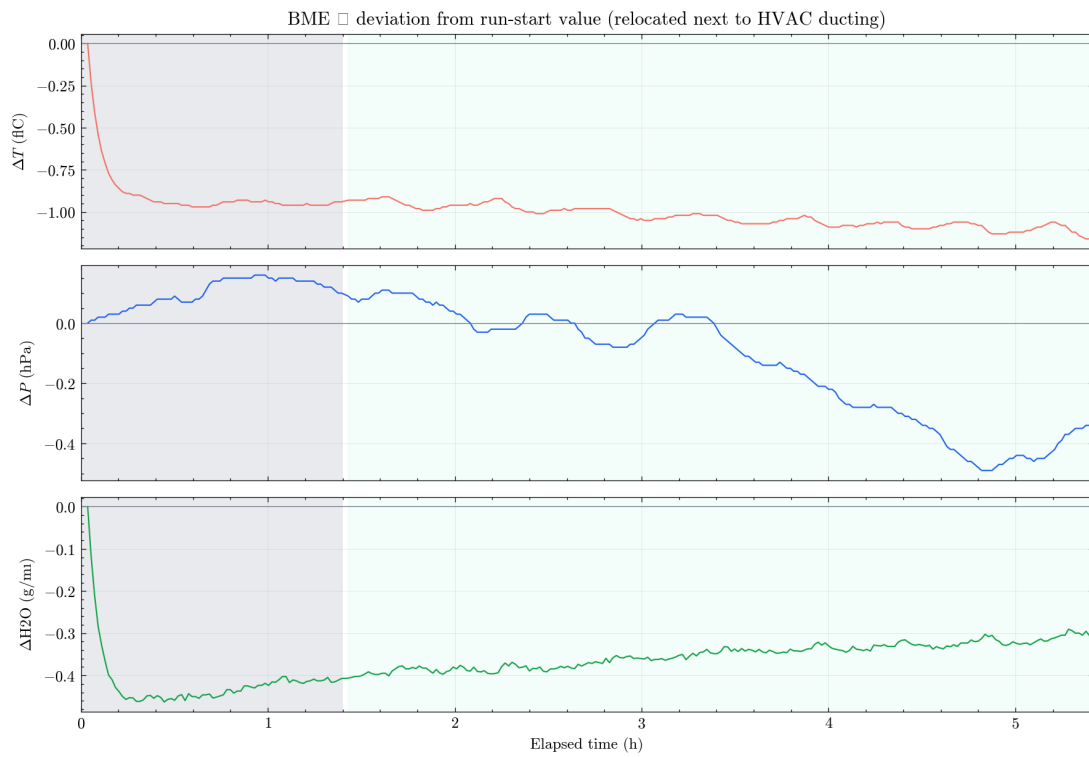


Figure 15: BME T/P/H₂O as deviation from run-start value.

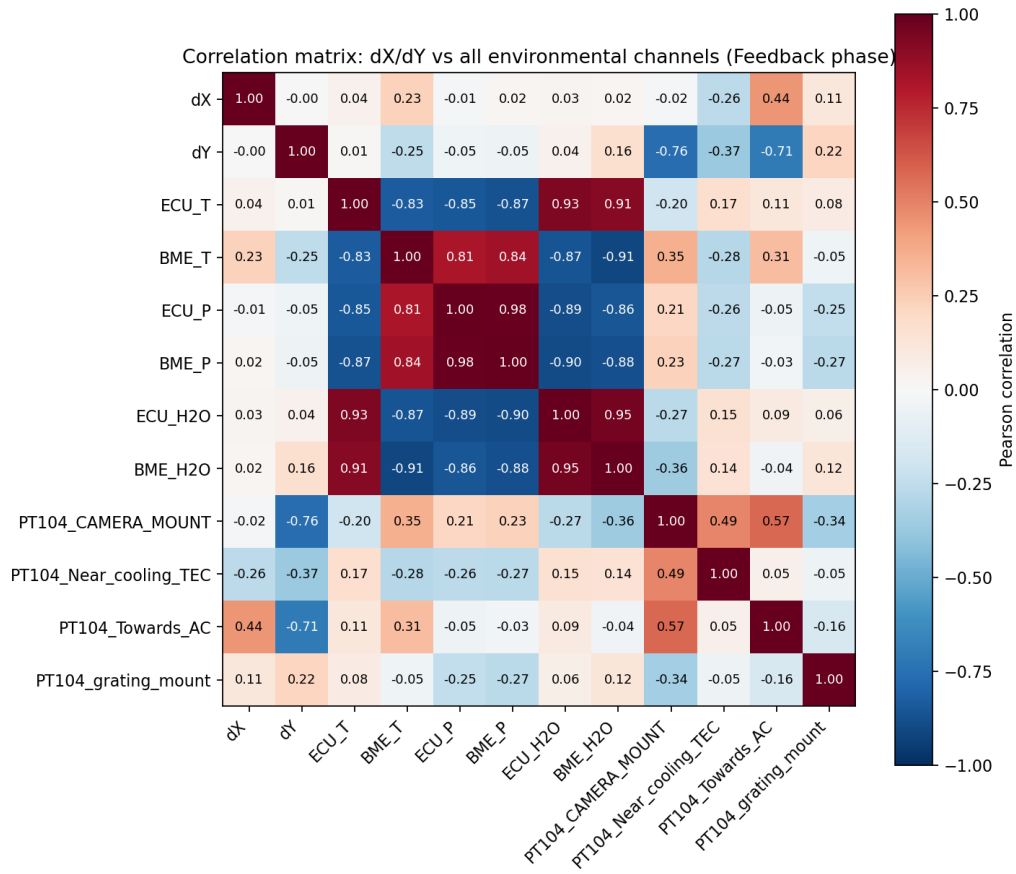


Figure 16: Feedback-phase Pearson correlation, all sensor channels including the 4 new PT104 channels.